

Assignment # 2

Student Task Management System

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Project Overview

The Student Task Management System is a productivity-focused application designed to help students efficiently organize and track their academic tasks. The system enables students to create tasks, assign priority levels, and manage task statuses such as Pending, In Progress, and Completed, allowing for clear visibility of workload and progress. The primary goal of the application is to improve time management, task prioritization, and overall academic productivity.

Project Idea

The core idea behind the Student Task Management System is to move beyond traditional to-do lists that lack structure, prioritization, and progress tracking. Instead of simple task entries, the system introduces a well-organized task workflow where students can create tasks, assign priority levels, and update their status as Pending, In Progress, or Completed. This structured approach helps students better understand their workload and manage time more effectively.

By providing clear visual indicators for task priority and completion status, the application encourages consistent task monitoring and timely completion. The intuitive and minimal interface ensures ease of use, while real-time updates allow students to track progress at a glance. Overall, the system aims to transform task management into a streamlined and motivating experience, helping students stay organized, focused, and productive throughout their academic journey.

Main Features

Authentication System

Student Management System includes a secure and reliable authentication system that allows users to create accounts and log in safely. User credentials are protected through proper password hashing techniques to prevent unauthorized access. Session management mechanisms ensure that users remain logged in across app sessions without compromising security. This authentication layer provides a personalized experience by linking progress, XP, and performance statistics to individual user accounts.

Task Creation and Management

The system allows students to create, update, and delete tasks with ease. Each task includes essential details such as task title, description, and due date. This feature helps students organize academic responsibilities in a structured manner and ensures that all

tasks are clearly defined and easy to manage.

Task Priority System

To help students focus on important activities, the application supports task prioritization. Students can assign priority levels such as High, Medium, or Low to each task. Visual indicators are used to distinguish priorities, allowing users to quickly identify urgent tasks and manage their time more effectively.

Task Status Tracking

The task status feature enables students to track progress by categorizing tasks as Pending, In Progress, or Completed. This clear workflow helps users monitor ongoing activities and maintain motivation by visually recognizing completed tasks. Status updates occur in real time, providing an accurate overview of current workload.

Database Design and Functionality

Student Task Management System adopts a local-first database strategy to ensure fast performance and meaningful offline functionality. This approach allows users to access their data and continue interacting with the application even without an active internet connection. By prioritizing local storage, the application reduces dependency on continuous network availability while maintaining reliable data persistence.

Database Technologies

For mobile platforms including Android and iOS, the app utilizes SQLite through the expo-sqlite library. SQLite provides a lightweight, relational database solution suitable for mobile applications. For web compatibility, localforage is used as a fallback storage mechanism, ensuring consistent behavior across platforms.

Database Schema

The primary database table in Trivi is the Users table. It stores essential user-related information including a unique user ID as the primary key, username, email address, securely hashed password, and accumulated XP. This schema supports efficient authentication, progress tracking, and leaderboard calculations.

Database Functionality

The database is responsible for persisting user credentials and progress data locally. It automatically initializes when the application starts, creating required tables if they do not already exist. This ensures data integrity and smooth application operation while providing reliable offline access to user information.

Unique Feature: Structured Task Workflow System

The most distinctive feature of the Student Task Management System is its structured task workflow that focuses on clarity, prioritization, and progress tracking. Unlike basic to-do list applications that only allow task entry, this system provides a well-defined lifecycle for each task, enabling students to systematically manage their academic responsibilities from creation to completion.

Priority-Based Task Organization

Each task can be assigned a priority level such as High, Medium, or Low, allowing students to identify urgent and important tasks at a glance. This prioritization helps students allocate time effectively and focus on critical academic work without feeling overwhelmed.

Status-Driven Progress Tracking

The application enforces clear task statuses including Pending, In Progress, and Completed. This status-driven approach enables students to track real-time progress and maintain accountability. Visually separating tasks based on their current state helps users stay motivated and organized.

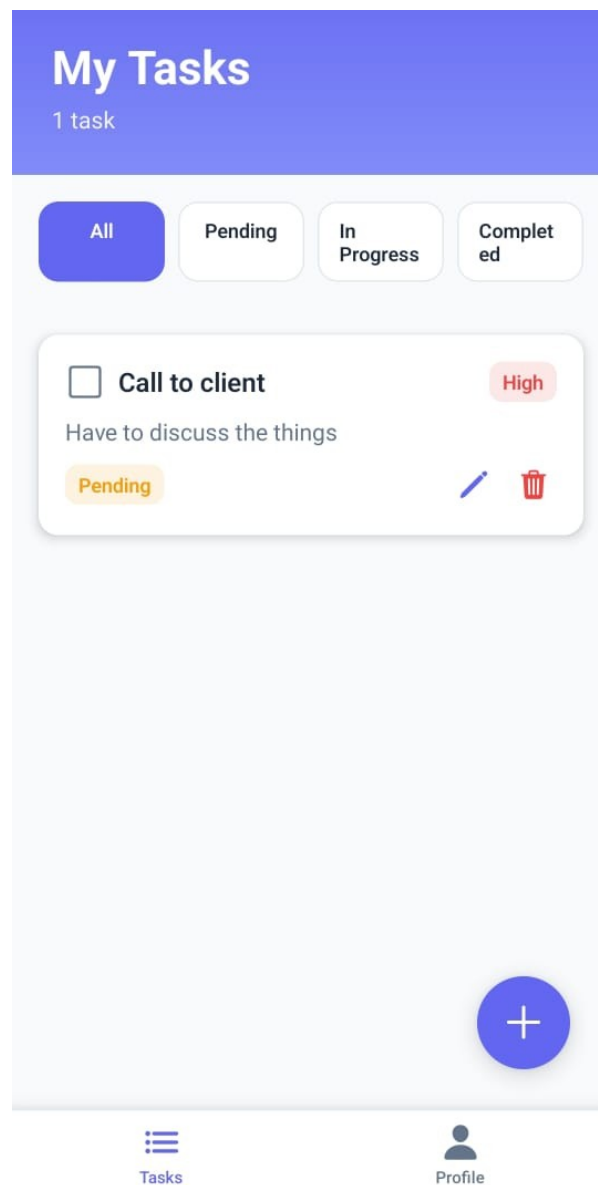
Consistent Task Structuring

All tasks follow a standardized structure that includes essential details such as title, description, priority, and status. This consistency ensures reliable task handling throughout the system.

Screenshots

User Dashboard

This subsection includes screenshots of the user dashboard, which serves as the central control panel of the Student Task Management System. The dashboard provides students with a comprehensive overview of their tasks, including tasks categorized by priority and status (Pending, In Progress, and Completed). It allows users to quickly assess their workload and track progress at a glance.



Authentication Screens

This subsection includes screenshots of the login and signup interfaces of the application. These screens demonstrate the secure authentication flow, user-friendly input fields, and overall visual consistency of the application's design.

Create Account

Start managing your tasks today

Username

Choose a username

Email

Enter your email

Password

Create a password

Confirm Password

Confirm your password

Register

Already have an account? [Login](#)

Welcome Back!

Login to manage your tasks

Email

Enter your email

Password

Enter your password

Login

Don't have an account? [Register](#)

User Profile Dashboard

This subsection displays screenshots of the user profile dashboard, showing key user information such as task status, completion rate, and performance statistics. It demonstrates how user progress is monitored and visually communicated.

